J1 RASPBERRY PI 4 MODEL B — 40-PIN GPIO HEADER

2×20, 2.54 mm • pin names = BCM • all I/O 3.3 V, NOT 5 V tolerant

3V3	1	+3V3	W01		
5V	2	+5V — DO NOT CONNECT			
GPIO2 / SDA1	3	I2C_SDA	W03		
5V	4	+5V — DO NOT CONNECT			
GPIO3 / SCL1	5	I2C_SCL	W04		
GND	6		W05	GND — common reference	
GPIO4 / GPCLK0	7	NC — freed in Rev B (was ENABLE)			
GPIO14 / TXD0	8				
GND	9				
GPIO15 / RXD0	10				
GPIO17	11	TACH_IN	W02		
GPIO18 / PCM_CLK	12	PH_A+ (J1-12 → U2-2)	W07	2	1A
GPIO27	13				
GND	14				
GPIO22	15				
GPIO23	16	PH_A− (J1-16 → U2-7)	W08	7	2A
3V3	17				
GPIO24	18	PH_B+ (J1-18 → U2-10)	W09	10	3A
GPIO10 / MOSI	19				
GND	20				
GPIO9 / MISO	21				
GPIO25	22	PH_B− (J1-22 → U2-15)	W10	15	4A
GPIO11 / SCLK	23				
GPIO8 / CE0	24				
GND	25				
GPIO7 / CE1	26				
GPIO0 / ID_SD	27				
GPIO1 / ID_SC	28				
GPIO5	29				
GND	30				
GPIO6	31				
GPIO12 / PWM0	32				
GPIO13 / PWM1	33				
GND	34				
GPIO19	35				
GPIO16	36				
GPIO26	37	ENABLE → R1 220 Ω ¼ W → EN_LED	W11 / W12		
GPIO20	38				
GND	39				
GPIO21	40				

U3 AS5600 MAGNETIC ROTARY ENCODER

12-bit, I²C addr 0x36, 3.3 V supply

VCC	W01	GND	W28 → GND (DIR = GND → CW increments)
SDA	W03	DIR	W29 → GND
SCL	W04	OUT	NC (analog/PWM out unused)

U4 IR SPEED SENSOR (TACH)

digital, 2 pulses / rev

VCC	W02	GND	W30 → GND
OUT	W06		

U1 4N25 OPTOCOUPLER (DIP-6)

HW enable isolation / quiet-start

1	A	B	6 NC (from PS1 CH1 (VCC1 rail))
2	K	C	5 → W14 +5V
3	NC	E	4 → DRV_EN → W15 → W16 → R2 10 kΩ pull-down → W17 → GND (returns to J1-6)

R1 220 Ω ¼ W — *I_F ≈ (3.3 − 1.2) / 220 ≈ 9.5 mA*

PS1 BENCH DC POWER SUPPLY (dual output)

Set current limit BEFORE enabling outputs

- W20 → +5V (VCC1) → +5.0 V (CH1)
- W21 → +12V (VCC2) → +12.0 V (CH2)
- W22 → GND → COM / GND
- VCC1/VCC2 come ONLY from PS1 — never from the Pi 5V pins (2/4)

U2 SN754410NE

Quad half-H driver, DIP-16, 4.5–36 V, 1 A/ch

16	VCC1	logic 5 V		
8	VCC2	motor 12 V		
1A	3	1Y	COIL_A1	W24 → A1 / Lead 1
2A	6	2Y	COIL_A2	W25 → A2 / Lead 2
3A	11	3Y	COIL_B1	W26 → B1 / Lead 3
4A	14	4Y	COIL_B2	W27 → B2 / Lead 4
1,2EN	DRV_EN 1	W18		
3,4EN	DRV_EN 9	W19		
GND	GND	GND	GND	4 / 5 / 12 / 13 → W23 → GND

H-BRIDGE A — drivers 1 & 2 → coil A

H-BRIDGE B — drivers 3 & 4 → coil B

EN HIGH = outputs active
EN LOW = outputs Hi-Z

pins 4/5/12/13 = GND + heatsink

M1 NEMA 17 BIPOLAR STEPPER

4-wire, 2 phases, 200 steps/rev (1.8°)

- COIL A: A1 / Lead 1, A2 / Lead 2
- COIL B: B1 / Lead 3, B2 / Lead 4
- M — rotor

12 V switched, up to ~1 A per phase

Shaft: diametric magnet → U3
Slotted/marked disc → U4

ENABLE PATH LOGIC (Rev B)

GPIO26	U1 LED	DRV_EN	U2 out
LOW / boot	OFF	0 V (R2)	Hi-Z
float / reset	OFF	0 V (R2)	Hi-Z
HIGH 3.3 V	ON ~9.5 mA	≈ +5 V	ACTIVE

GPIO26 default = internal pull-DOWN at power-up (GPIO4 defaulted to pull-UP). Firmware: ENABLE_PIN = 26

SCHEMATIC NOTES

1. Wire tags W01–W30 match the wire list on sheet 2.
2. Junction dot = electrical connection. A wire hop (arc) or a plain crossing without a dot = NO connection.
3. All ground symbols are one common net: J1-6, PS1 COM, U2-4/5/12/13, U1-2, R2, U3 GND & DIR, U4 GND.
4. +5V and +12V are sourced ONLY by PS1. No 5 V or 12 V signal may touch any J1 pin (3.3 V logic only).
5. U2-1 (1,2EN) and U2-9 (3,4EN) are tied: net DRV_EN.
6. R2 holds DRV_EN at 0 V until GPIO26 is driven HIGH.
7. Orange + Δ B = changed in Rev B (enable GPIO4 → GPIO26, J1-7 → J1-37). All other connections unchanged from Rev A.
8. J1 pins shown × are unused (no connection).
9. Pin numbers shown at the component body (grey, mono).
10. Pi provides on-board 1.8 kΩ pull-ups on SDA1/SCL1.
11. Set PS1 current limit first; power the Pi and verify DRV_EN = 0 V before enabling PS1 outputs.

LEGEND

- 3.3 V logic signal
- +5 V / +3V3 rail
- +12 V / motor
- Rev B enable path
- junction
- hop = no connect
- unused / NC
- W07 wire ID
- GND

PARMCO — STEPPER CONTROL HARDWARE SCHEMATIC

PROJECT: ECSE 4235 Final Project — PARMCO		SIZE / SCALE: ANSI B • NTS
DOCUMENT NO.: SCH-RP4STEP-V1	REV: B	SHEET: 1 of 2
DRAWN BY: Macey Glad	DATE: 27 Oct 2026	SUPERSEDES: Rev A (GPIO4 EN)

REVISION HISTORY

A 19 Oct 2026 Initial release (driver enable on GPIO4 / J1-7)
B 19 Oct 2026 Enable moved to GPIO26 / J1-37 — redrawn 27 Oct 2026

PARMCO HARDWARE — WIRE LIST, NETLIST & PIN CROSS-REFERENCE (Rev B)

Every wire tag (Wxx) on sheet 1 is listed below with both endpoints, net name, signal class and expected level. Orange rows changed in Rev B (driver enable moved GPIO4 / J1-7 → GPIO26 / J1-37).

1. WIRE LIST (point-to-point)

Wire	From (pin)	To (pin)	Net	Signal class	Level	Notes
W01	J1-1 (3V3)	U3 VCC	+3V3	Power	3.3 V	Encoder supply
W02	+3V3 junction (W01)	U4 VCC	+3V3	Power	3.3 V	IR sensor supply (hops over SDA/SCL)
W03	J1-3 (GPIO2 / SDA1)	U3 SDA	I2C_SDA	I²C data, bi-dir	3.3 V	Angle read, addr 0x36
W04	J1-5 (GPIO3 / SCL1)	U3 SCL	I2C_SCL	I²C clock	3.3 V	Pi on-board 1.8 kΩ pull-ups
W05	J1-6 (GND)	GND net	GND	Ground	0 V	Pi ground reference to common GND
W06	J1-11 (GPIO17)	U4 OUT	TACH_IN	Digital in	3.3 V	2 pulses / rev
W07	J1-12 (GPIO18)	U2-2 (1A)	PH_A+	Digital out	3.3 V	Driver 1 input, coil A +
W08	J1-16 (GPIO23)	U2-7 (2A)	PH_A−	Digital out	3.3 V	Driver 2 input, coil A −
W09	J1-18 (GPIO24)	U2-10 (3A)	PH_B+	Digital out	3.3 V	Driver 3 input, coil B +
W10	J1-22 (GPIO25)	U2-15 (4A)	PH_B−	Digital out	3.3 V	Driver 4 input, coil B −
W11	J1-37 (GPIO26)	R1 pin 1	ENABLE	Digital out	3.3 V	REV B: was J1-7 (GPIO4)
W12	R1 pin 2	U1-1 (anode)	EN_LED	LED drive	≈1.2 V	I_F ≈ 9.5 mA when HIGH
W13	U1-2 (cathode)	GND net	GND	Ground	0 V	LED return to J1-6 via common GND
W14	+5V (PS1 CH1)	U1-5 (collector)	+5V	Power	5.0 V	Opto switched-high rail
W15	U1-4 (emitter)	DRV_EN junction	DRV_EN	Enable	0 / ≈5 V	Opto output
W16	DRV_EN junction	R2 pin 1	DRV_EN	Enable	0 / ≈5 V	10 kΩ pull-down
W17	R2 pin 2	GND net	GND	Ground	0 V	Holds DRV_EN LOW at boot
W18	DRV_EN	U2-1 (1,2EN)	DRV_EN	Enable	0 / ≈5 V	Bridge A enable
W19	DRV_EN	U2-9 (3,4EN)	DRV_EN	Enable	0 / ≈5 V	Bridge B enable (tied to U2-1)
W20	PS1 +5.0 V (CH1)	U2-16 (VCC1)	+5V	Power	5.0 V	Driver logic supply — NOT from Pi
W21	PS1 +12.0 V (CH2)	U2-8 (VCC2)	+12V	Power	12.0 V	Motor supply — NOT from Pi
W22	PS1 COM	GND net	GND	Ground	0 V	Supply return
W23	U2-4 / 5 / 12 / 13	GND net	GND	Ground / thermal	0 V	All four pins tied, heatsink
W24	U2-3 (1Y)	M1 A1 (lead 1)	COIL_A1	Motor drive	0 / 12 V	Phase A
W25	U2-6 (2Y)	M1 A2 (lead 2)	COIL_A2	Motor drive	0 / 12 V	Phase A
W26	U2-11 (3Y)	M1 B1 (lead 3)	COIL_B1	Motor drive	0 / 12 V	Phase B
W27	U2-14 (4Y)	M1 B2 (lead 4)	COIL_B2	Motor drive	0 / 12 V	Phase B
W28	U3 GND	GND net	GND	Ground	0 V	Encoder ground
W29	U3 DIR	GND net	GND	Config	0 V	CW = increasing angle
W30	U4 GND	GND net	GND	Ground	0 V	Sensor ground

2. NETLIST

Net	Nodes (component-pin)	Notes
+3V3	J1-1, U3 VCC, U4 VCC	3.3 V from Pi regulator (sensors only)
+5V	PS1 CH1, U2-16 (VCC1), U1-5	Bench supply only
+12V	PS1 CH2, U2-8 (VCC2)	Bench supply only
GND	J1-6, PS1 COM, U1-2, R2-2, U2-4/5/12/13, U3 GND, U3 DIR, U4 GND	Single common ground
I2C_SDA	J1-3, U3 SDA	
I2C_SCL	J1-5, U3 SCL	
TACH_IN	J1-11, U4 OUT	
PH_A+	J1-12, U2-2	
PH_A−	J1-16, U2-7	
PH_B+	J1-18, U2-10	
PH_B−	J1-22, U2-15	
ENABLE	J1-37, R1-1	Rev B
EN_LED	R1-2, U1-1	Rev B path
DRV_EN	U1-4, R2-1, U2-1, U2-9	Default LOW via R2
COIL_A1	U2-3, M1 A1	
COIL_A2	U2-6, M1 A2	
COIL_B1	U2-11, M1 B1	
COIL_B2	U2-14, M1 B2	
(no connect)	J1-7 (freed in Rev B), U1-3, U1-6, U3 OUT; all other J1 pins	J1-2/4 (+5V) must stay open

3. REFERENCE DESIGNATORS

Ref	Description	Qty
J1	Raspberry Pi 4 Model B, 40-pin header	1
U1	4N25 optocoupler, DIP-6	1
U2	SN754410NE quad half-H driver, DIP-16	1
U3	AS5600 magnetic encoder module	1
U4	IR speed sensor module (digital out)	1
R1	220 Ω, ¼ W	1
R2	10 kΩ, ¼ W (pull-down)	1
M1	NEMA 17 bipolar stepper, 4-wire	1
PS1	Bench DC supply, 5 V + 12 V, current-limited	1

4. PRE-POWER CONTINUITY CHECKS (Rev B)

#	Check	Probe points	Pass
1	Supply isolation	U2-16 and U2-8 to every J1 pin	Open circuit
2	Common ground	J1-6 ↔ PS1 COM ↔ U2-4/5/12/13 ↔ U1-2 ↔ U3/U4 GND	≈ 0 Ω
3	Enable pull-down	U1-4 (DRV_EN) to GND	10 kΩ ± 5 %
4	Enable pin move	J1-37 → U1-1; confirm J1-7 open	≈ 220 Ω; open
5	EN tie	U2-1 ↔ U2-9	≈ 0 Ω
6	Supply setpoints	PS1 CH1 / CH2, outputs off, limit on	5.0 / 12.0 V ± 0.2 V

Firmware: ENABLE_PIN = 26 (BCM). Drive LOW at init; set HIGH only after GPIO18/23/24/25 are outputs.

PARMCO — STEPPER CONTROL WIRE LIST & NETLIST

PROJECT: **ECSE 4235 Final Project — PARMCO**		SIZE / SCALE: **ANSI B • NTS**	
DOCUMENT NO.: **SCH-RP4STEP-V1**		REV: **B**	SHEET: **2 of 2**
DRAWN BY: **Macey Glad**	DATE: **27 Oct 2026**	SUPERSEDES: **Rev A (GPIO4 EN)**	

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